

The original cybersecurity dataset presented by Almahmoud (2023) contained 142 temporal nodes (data points) with 143 columns (features), covering the period from July 2011 to December 2022. Through systematic augmentation and temporal extension, this dataset has been substantially enhanced to 1,292 nodes (after smoothing) with 2,126 columns, now spanning July 2011 to December 2024. This represents an 810% increase in temporal observations and a 1,385% increase in features, significantly enriching the dataset's analytical depth and breadth.

Dimension	Original (Almahmoud 2023)	Augmented Dataset	Increase
Temporal Nodes	142	1,292 (smoothed)	+810%
Features (Columns)	143	2,126	+1,385%
Temporal Coverage	Jul 2011 - Dec 2022	Jul 2011 - Dec 2024	+24 months
Data Sources	5	11	+120%
Country Coverage	37	37 (maintained)	Maintained
Attack Taxonomies	2 (Hackmageddon, Elsevier)	6 (+ UMD, NVD, RSS, Dark Web)	+200%
Defence Taxonomies	1 (Elsevier PATs)	3 (+ RSS, Google Trends)	+200%

Quantitative Summary of the Augmentation

Data Source	Fields Added	Temporal Span	Key Contribution
Hackmageddon Extension	1,036 (extended)	2023-2024	Incident continuity
Elsevier Extension	147 (extended)	2023-2024	Academic discourse
Python Holidays Extension	38 (extended)	2023-2024	Temporal context
RSS Feeds	161	2011-2024	Media perspective
Google Trends	86	2011-2024	Public awareness
Bluesky	37	2023-2024	Conflict tracking
University of Maryland	444	2014-2024	Technical taxonomy
ACLED	37	2017-2024	Political violence
NVD/CVE	40	2011-2024	Vulnerability linkage
Dark Web Counts	67	2020-2024	Underground economy
Dark Web Pricing	68	2020-2024	Threat commoditisation
Total New/Extended	2,161		

Feature Breakdown by Source

Hackmageddon (2011-2024)

The extension of the Hackmageddon data set by 24 months (2023 –2024) captures the evolving threat landscape through 2023-2024. This would potentially include increased ransomware sophistication and prevalence, effects of geopolitical cyber conflicts (e.g. Russia-Ukraine and Middle East tensions), supply chain attack evolution and post-pandemic attack pattern changes.

Twitter/Bluesky (2011-2024)

Due to Twitter's API (used in Almahmoud (2023)) access restrictions and platform change post-2022 that limited research access an alternative data source was necessary for global conflict discourse monitoring. Bluesky emerged as viable alternative, due to the decentralised architecture and growing adoption by journalists, researchers, and policy analysts, enabling quality discourse. The collection script, which collects data for the period of January 2023 to December 2024, maintains methodological consistency in conflict mention extraction and country attribution to preserve prediction stability. The script collects counts of the following conflict related key terms, by country and by month.

```
for term in clean_terms:
    if len(term) > 2: # Avoid very short terms
        search_combinations.extend([
            f"{term} war",
            f"{term} conflict",
            f"{term} military",
            f"{term} army",
            f"{term} attack",
            f"{term} defense"
        ])
    )
```

This integration is valuable because it allows for the testing of whether geopolitical tensions correlate with state-sponsored cyber operations, whether conflict discourse indicates heightened cyber warfare risk and whether cyber-attacks increasingly integrated into hybrid warfare strategies. In turn, exploring these potential connections provides valuable and actionable insights into how geopolitics can affect cyber security, both on a national and corporate level.

RSS feed scraping (2011 – 2024)

A comprehensive retrospective collection of 138 cybersecurity-focused RSS feeds was implemented, including both historical and current feeds from Vendor Blogs, Security News Outlets, Research Organisations and Vulnerability Databases (using the feedparser library). The script compiled an NLP pipeline for attack type identification, keyword and semantic matching across 42 attack categories and a monthly aggregation of mention frequencies for extracting attack type mentions. Similarly, for solutions, a parallel NLP pipeline for PATs discussion tracking was implemented for semantic extraction across 119 categories to capture technology adoption discourse, implementation challenges, and efficacy discussions.

This addition enriches the dataset for several reasons. First, it relates a media and industry perspective by capturing industry reporting distinct from academic (Elsevier) or incident databases (Hackmageddon). Again, RSS feeds reflect immediate security community response (real time discourse), unlike lagged academic publications and this emphasises actual operational security concerns and practitioner priorities, ensuring a practical focus.

Google Trends Worldwide Topic Data (2011 –2024)

Google Trends queries with "Worldwide" geographic scope are used for Global Search Interest Tracking. The data was downloaded directly from Google News. The normalised interest scores (0-100 scale), where 100 means peak search interest, are collected and stored in a csv file, covering July 2011 to

December 2024 with monthly granularity. “Topic” rather than “Search Term” results are used to increase semantic breadth. 29 attacks related and 57 PATs related search terms were collected and analysed.

This data is particularly valuable for several reasons. First, public awareness data is captured where search volume indicates societal concern, threat salience and PAT popularity. Also, the trends may correlate with cyber security incidents tracked by other fields, so we might expect to see spikes following major attacks (e.g., WannaCry in May 2017). We can also track seasonal patterns to identify recurring temporal patterns in security awareness and assess correlations with attacks launched. Moreover, the data may serve as a predictive indicator, as search interest may precede or follow actual attack volumes. However, it is important to note that there is a regional blind spot, as the data only reflects primarily English-language, Western search behaviour

University of Maryland Cyber Event Database (2014-2024)

The integration of the University of Maryland’s Cyber Event Database (University of Maryland's Centre for the Study of Terrorism and Responses to Terrorism (START)) allowed for the addition of a comprehensive threat taxonomy, covering 132 months (January 2014 to December 2024). The attacks are tracked against the common 37 country taxonomy with the addition of unknown locations and an aggregate count, creating 444 new fields.

The database tracks 12 distinct technical attack methodologies:

1. Exploitation of Application Server (indicating web application compromises)
2. Data Attack (indicating integrity or availability targeting)
3. External Denial of Service (indicating DDoS from external networks)
4. Exploitation of End Hosts (indicating user device compromises)
5. Message Manipulation (indicating communication tampering)
6. Undetermined (meaning there is an unclear methodology)
7. Exploitation of Sensors (IoT/sensor targeting)
8. Exploitation of Network Infrastructure (Router/switch attacks)
9. Internal Denial of Service (Internal resource exhaustion)
10. Physical Attack (Hardware tampering)
11. Exploitation of Data in Transit (Network traffic interception)
12. Exploitation of Network Server (Server-level compromises)

The database tracks an aggregation of publicly reported cyber events, which undergo expert classification by trained analysts using technical indicators and are then cross-referenced with open-source intelligence (OSINT). This database is distinct from Hackmageddon because the Maryland database emphasises attack methodology rather than malware family so provides richer technical context per incident. As such, the taxonomies are complementary because combining sources enables multi-faceted attack characterisation. The integration of the Maryland database is particularly valuable because the Hackmageddon incident counts can now be validated through independent confirmation. Additionally, the supporting taxonomy enables the analysis of attack methodology evolution distinct from malware trends.

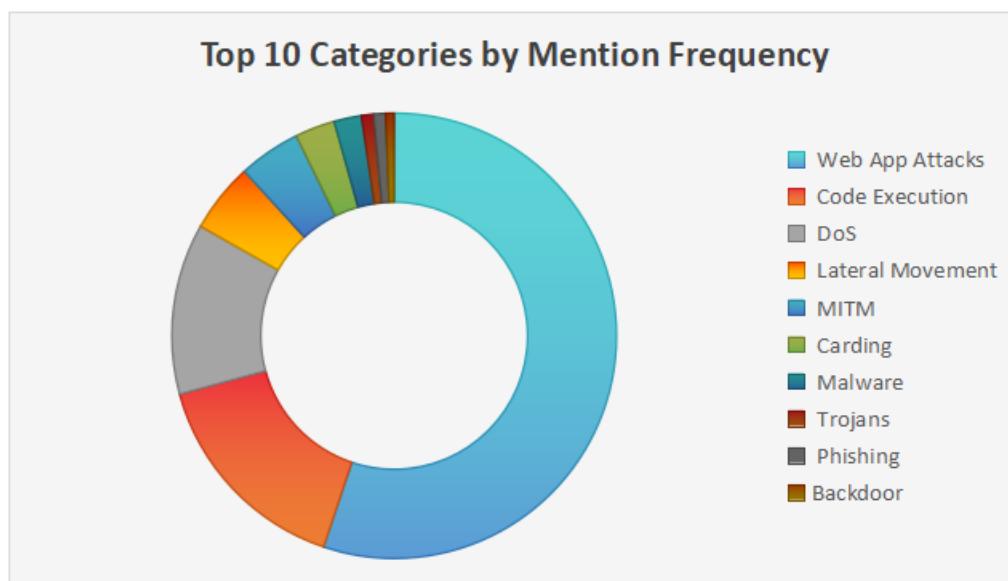
ACLED Political Violence Data (2017-2024)

The ACLED (Armed Conflict Location & Event Data Project) Political Violence Database (covering 96 months) allows for the granular tracking of conflict events, owing the real-time global conflict and crisis

data captured. So, the aggregation of monthly event counts by country is valuable because it enables a "ground truth" on conflict data, whilst Twitter and BlueSky capture the information environment. This, in turn, allows for the empirical analysis of whether there is a correlation between kinetic conflict and cyber operations, whether there is an increase in state-sponsored cyber activity during military engagements, whether cyber operations ought to be considered as conflict precursors or escalation indicators through the analysis of geographic clustering of physical and cyber warfare.

National Vulnerability Database (NVD/CVE) Data (2011-2024)

The integration of NVD data provides comprehensive vulnerability intelligence from July 2011 to December 2024. From the CVE descriptions, a count of aggregated vulnerability mentions for each month was collected. NLP techniques were applied to the CVE description fields, enabling keyword and semantic extraction for 40 attack categories and then monthly aggregation of mention frequencies was calculated. This has distinct analytical value because it allows for the connection of disclosed weaknesses to active threat landscape (Vulnerability Exploit Linkage); it can enable defence prioritisation by identifying which vulnerability types enable which attack classes; and because temporal analysis can track the lag between vulnerability disclosure and exploitation.



Dark Web Marketplace Data (2020-2024)

A major enhancement to the original dataset is the integration of Dark Web Marketplace Data. The dataset used for this project captures underground economy intelligence from 2020 to 2024, by scraping 14 major dark web marketplace platforms and the connected discussion boards for each marketplace where actors share techniques, tools, and services for mention counts and listed prices of 67 attack categories.

- Darkode, a hacking forum
- Tor2door, a general marketplace with strong cyber tools section
- Bohemia
- Nemesis, containing exploit listings
- Royal Market, a ransomware-as-a-Service hub

- Vice City, a longstanding marketplace
- ASAP Market, "As Safe As Possible" marketplace
- MGM Grand Market, a casino-themed marketplace
- AlphaBay Market
- Dark Market
- Torrez
- Whitehouse Market
- Abacus
- Dark Matter, an emerging from 2023-2024 platform

The data collection pipeline uses the Tor browser automation for marketplace access, employs ethical scraping protocols (e.g. no transaction facilitation, no law enforcement action interference) and collects pseudonymised data collection of both attack price data and attack mention frequency.

To account for the gap between dark web dialect and academic language, a mention list specialised for the dark web and a mapping of this mention list to target terms was developed. The attack categories correspond to the mention lists below.

```
mention_list = [
    # Initial Access
    ('Phishing', 'Spear Phishing', 'Whaling', 'BEC', 'Email Spoofing'),
    ('Social Engineering', 'Pretexting', 'Baiting', 'Vishing', 'Smishing'),
    ('Email Bombing', 'Mailbomb', 'Spam Attack'),
    ('Drive-by Download', 'Watering Hole Attack'),
    ('Exploit Kit', 'Zero-day', '0-day'),
    # Malware
    ('Malware', 'Malicious Software'),
    ('Ransomware', 'Crypto-ransomware', 'Locker Ransomware'),
    ('Ransomware-as-a-Service', 'RaaS'),
    ('Trojan', 'Banking Trojan', 'Dropper', 'Downloader'),
    ('Backdoor', 'RAT', 'Web Shell'),
    ('Info-Stealer', 'Spyware', 'Keylogger'),
    ('Worm', 'Virus', 'Self-propagating Malware'),
    ('Botnet', 'C2 Server', 'Command and Control'),
    ('Rootkit', 'Bootkit'),
    ('Adware', 'Malvertising', 'PUPs'),
    ('Wiper Malware', 'Data Destruction Malware'),
    ('Malware-as-a-Service', 'MaaS', 'CaaS'),
    # Network & App Attacks
    ('Denial-of-Service', 'DoS', 'DDoS'),
    ('SQL Injection', 'SQLi'),
    ('Cross-site Scripting', 'XSS'),
    ('Remote Code Execution', 'RCE'),
    ('Man-in-the-Middle', 'MITM', 'Session Hijacking'),
    ('DNS Spoofing', 'DNS Hijacking', 'Pharming'),
    ('Password Attack', 'Credential Stuffing', 'Brute Force'),
    # Monetisation & Fraud
```

```
('Carding', 'Card Fraud', 'Fullz', 'Card Dump', 'CVV'),  
( 'Account Takeover', 'ATO', 'Bank Log'),  
( 'Cryptojacking', 'Illegal Cryptomining'),  
( 'Data Breach', 'Data Leak', 'Data Exfiltration'),  
( 'Dark Web Market', 'Underground Market', 'Carding Forum'),
```

Advanced Tactics

```
('Lateral Movement', 'Privilege Escalation', 'Persistence'),  
( 'Advanced Persistent Threat', 'APT'),  
( 'Supply Chain Attack', 'Third-party Attack'),  
( 'Insider Threat', 'Malicious Insider'),  
( 'IoT Attack', 'OT Attack', 'ICS Attack'),  
( 'Adversarial Attack', 'AI Poisoning', 'Data Poisoning'),
```

]

Price data for these attack types is also extracted. Prices are converted into USD conversion from cryptocurrencies using monthly average exchange rates. Then, price averaging takes place across multiple concurrent listings, with outlier removal (which could be honeypots or scams). It is important to note that the data collected was collected for research purposes only.

The value of this augmentation is myriad. First, threat commoditisation tracking is possible as price decreases indicate capability democratisation and price spikes suggest increased demand or reduced supply (e.g. law enforcement action). Capability accessibility can also be inferred as low prices suggest a high risk of widespread exploitation. Crucially, this can enhance predictive intelligence as dark web activity precedes real-world attack campaigns by weeks/months.

Impact of the Novel Dataset

This new, augmented dataset allows for multi-perspective threat intelligence, enabling triangulation across six main areas. Incident reports (such as Hackmageddon, UMD), which answers, "What attacks occurred?". Academic research (represented by Elsevier) which provides insight into what is being studied. Media discourse (from RSS feeds), which informs what is being reported. Public awareness (from Google Trends) which indicates what is being searched. Underground activity (from the Dark Web) answering "What is being commoditised?". Finally, vulnerability disclosure (from NVD) assess what weaknesses are being discovered. Synchronising these investigations allows for a more holistic approach to cyber threat intelligence.

Additionally, the dataset provides longitudinal analysis capabilities. The 13.5-year timespan enables long-term trend identification, such as an analysis of cyber-attacks across generational shifts in technology (e.g. Web 2.0 to Cloud to IoT to AI eras), economic cycle correlation (e.g. 2008 crisis aftermath, COVID-19 pandemic, post-pandemic) and geopolitical event correlation (e.g. Arab Spring, Ukraine conflicts 2014 & 2022, etc).

In summation, the transformation of Almahmoud's (2023) foundational dataset from 142 nodes and 143 features to 1,292 nodes and 2,126 features represents not only quantitative expansion, but a paradigm shift from single-source analysis to multi-perspective predictive intelligence. The 14-fold increase in features and 8-fold increase in temporal observations transforms this from an academic dataset into a strategic intelligence platform supporting operational defence, policy development, and scientific discovery.

The extension to the end of 2024, maintains methodological continuity whilst adapting to platform disruptions (e.g. Twitter to Bluesky transition). The integration of dark web marketplace data allows for systematic economic analysis of cyber threat commoditisation through marketplace monitoring. The multi-source triangulation validates findings across incident reports, academic research, media coverage, public awareness, and underground activity. In keeping with the principles of open science, the GitHub repository includes comprehensive data provenance and releasing the maximum shareable data for community benefit, to allow more innovation.

This augmented dataset is part of several shifts in cyber security research. It is predictive rather than descriptive, proactive rather than reactive, quantitative rather than qualitative (owing to the economic modelling of cybercrime supply-demand dynamics capabilities), and integrated rather than siloed. These features together allow for a holistic understanding of cyber-physical-information threat nexus. The dramatic increase in nodes and features, from Almahmoud's (2023) work, represents not just significant data accumulation, but could be a part of quantitative, evidence-based shift in the discipline of cybersecurity research.